QUESTION BANK

BE-I Sem Sub: calculus and Linear Algebra Sub code: CSMAT101

Module-III

1. Find the rank of following matrices by reducing into row echelon form

$$\begin{bmatrix} 0 & 1 & -3 & 1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix} \quad \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} \quad \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$

2. Test for consistency and solve

$5x+3y+7z=4 \qquad x+y+z=6 \qquad x+2y+3z=14$

$3x+26y+2z=9 \qquad x-y+2z=5 \qquad 4x+5y+7z=35$

$7x+2y+10z=5 \qquad 3x+y+z=8 \qquad 3x+3y+4z=21$

3. Investigate the values of and such that the system of equation

$x+y+z=6,\ x+2y+3z=10,\ x+2y+\lambda z=\mu,$ *may have*

(a) *Unique solution* (b) *Infinite solution* (c) *No solution*

4. Solve the following system of equation using Gauss elimination and Gauss Jordan method

4.

$x+y+z=9 \qquad 2x+5y+7z=52 \qquad 2x_1+x_2+4x_3=12$

$x-2y+3z=8 \qquad 2x+y-z=0 \qquad 4x_1+11x_2-x_3=33$

$2x+y-z=3 \qquad x+y+z=9 \qquad 8x_1-3x_2+2x_3=20$

Module-IV

1. Prove that the set of C of all complex numbers (i.e. the set of all ordered pairs of real numbers) is vector space over the field R of all real numbers where vector addition is defined by $(x_1, x_2) + (y_1, y_2) = (x_1+y_1, x_2+y_2)$ for all $(x_1, x_2), (y_1, y_2) \in C$ and scalar multiplication is defined by $\alpha\,(x_1, x_2) = (\alpha x_1, \alpha x_2)$ for all $\alpha \in R$.
2. Show that all the set of all matrices of the type m x n where m and n are fixed positive integers is a vector space over R with respect to matrix addition and multiplication of matrix by a scalar.
3. Let V be the vector space of all square matrices over R determine which of the following are subspace of V

(i)$W = \left\{ \begin{bmatrix} x & y \\ z & 0 \end{bmatrix}; x, y, z \in R \right\}$ (ii)$W = \left\{ \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix}; x, y \in R \right\}$

(iii)W={A:A ∈ V and A is singular}

4. Let V be the vector space of all real valued continuous functions over R. show that the set W of solutions of differential equations $5\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 2y = 0$ is a subspace of V

5. Show that the vector (5,9,5) is a the linear combination of the vectors (2,1,4), (1,-1,3) and (3,2,5)
6. Show that the vector (2,2,3) is a the linear span of the vectors (2,1,4), (1,-1,3) and (3,2,5)
7. Find the values of 'k' for which the vector (1,-2,k) is a the linear combination of the vectors (3,0,-2) and (2,-1,5)
8. Show that the vector (5,9,5) is a the not linear span of S={(5,3,7), (3,26,2) , (7,2,10)}
9. Express the polynomial
   $f(x) = x^2 + 4x - 3$ as a linear combinatio n of $f_1(x) = x^2 - 2x + 5,\ f_2(x) = 2x^2 - 3x,\ f_3(x) = x + 3$
10. Decide whether or not the vector $M = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ in the vector second order squar matrices is a linear combination of the matrices $M_1 = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ $M_2 = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ and $M_3 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$
11. Define linear dependence and independence vectors. Check the following vectors for linear independence or dependence.
    $(i)\{(1,1,-1),(2,-3,5),(-2,1,4)\}$ $(ii)\{(1,3,2),(1,-7,-8),(2,1,-1)\}$
12. Show that the set of vectors S={91,1,2), (-3,1,0), (1,-1,1), (1,2,-3)} are linearly dependent
13. Given the set **S={(**0,2,3,4)(2,3,5,4)(4,8,13,12)} and T={(1,2,2,1),(2,1,1,2),(3,2,2,3),(0,1,1,0)} Show that both S and T are linearly dependent.